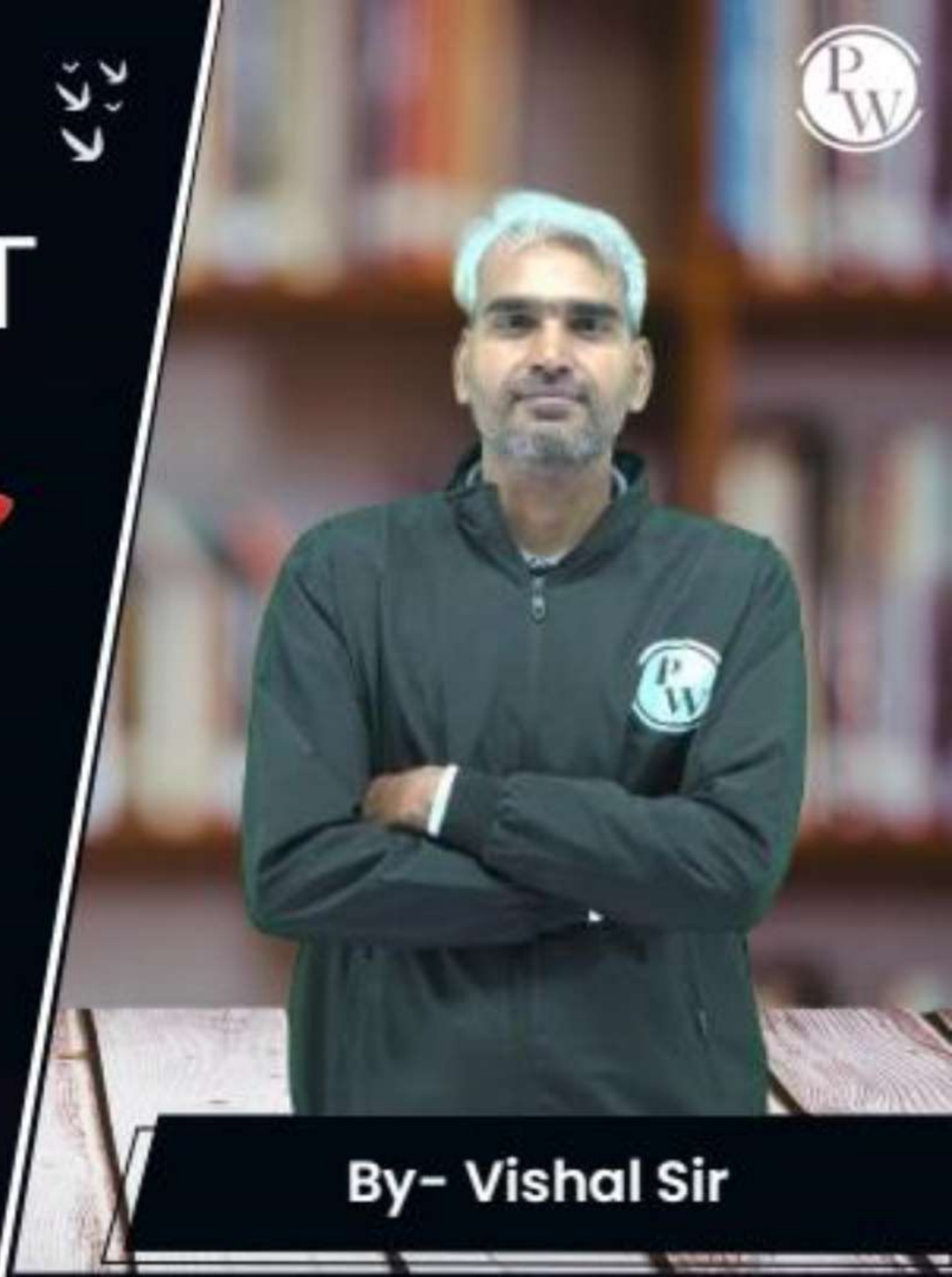
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Membership test
- **Topic** Relationship between two FD sets
- **Topic** FD set of a sub-relation

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Topics to be Covered



from last lecture

Today's TOPIC

- Topic: Minimal Cover (Canonical Cover)
- Topic: Procedure to obtain minimal cover for a given FD set
- Topic: Schema refinement (Normalization)
- Topic: Problems because of redundancy in a relation

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- Minimal Cover :- (or) Canonical Cover :-

Given a FD set F, minimal cover of FD set F is another FD set F_m, such that

(i) $F_m = F$

and (ii) F_m must be minimal

i.e. Nothing can be removed from F_m

i.e. F_m must not contain any redundant FD or extraneous attribute

#e.g., Consider the FD set

$F = \{A \rightarrow B,\ B \rightarrow C,\ A \rightarrow C\}$

↳ redundant FD, because it can be obtained from $A \rightarrow B$ & $B \rightarrow C$

∴ Minimal Cover {i.e. F_m} of F $= \{A \rightarrow B,\ B \rightarrow C\}$

#e.g., Consider the FD set

$F = \{AB \rightarrow C,\ A \rightarrow B\}$

$A \rightarrow B$

We know $A \rightarrow B$, ∴ If we have information of A then we don't need information about B explicitly

Hence

From $AB \rightarrow C$

this B can be removed {i.e. given $A \rightarrow B$, B is extaneous in $AB \rightarrow C$}

Minimal Cover of F {i.e, F_m} = $\{A \rightarrow C, A \rightarrow B\}$

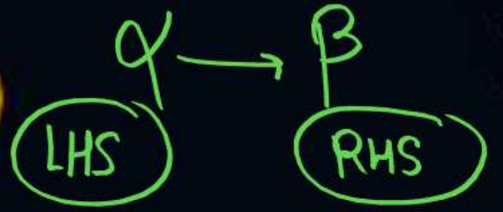


Topic : Minimal cover (Canonical cover)

- Minimal cover of canonical cover of FD set F is a set of functional dependencies (F_m)
  such that,
  - $F_m = F$ and
  - F_m does not contain any redundant FD, and F_m must not contain any extraneous attribute at either side of any of its FD



Topic : Testing if an Attribute is Extraneous



Consider a set F of functional dependencies and functional dependency $\alpha \rightarrow \beta$ in F.

To test if attribute $A \in \alpha$ is extraneous in α (i.e., Any extraneous attribute in LHS of FD)

1. compute $(\{\alpha\} - \{A\})^+$ using the dependencies in F.
2. check if $(\{\alpha\} - \{A\})^+$ contains A; if it does then, A is extraneous.

(otherwise A is not extraneous)

To test if attribute $B \in \beta$ is extraneous in β (i.e., Any extraneous attribute in RHS of FD)

1. compute α^+ using only the dependencies in F', $F' = (F - \{\alpha \rightarrow \beta\}) \cup \{\alpha \rightarrow (\beta - B)\}$.
2. check that α^+ contains B; if it does then, B is extraneous

(otherwise B is not extraneous)

#e.g., Consider the FD set

$F = \{A \to BC,\ B \to C\}$

- Check if B is extraneous in $A \to BC$

$(A)^+$ wrt $\{B \to C, A \to C\} = \{A, C\}$

$B \notin (A)^+$ wrt $\{B \to C, A \to C\}$

$\therefore$ B is not extraneous in $A \to BC$

- Check if C is extraneous in $A \to BC$

$(A)^+$ wrt $\{B \to C, A \to B\} = \{A, B, C\}$

$C \in (A)^+$ wrt $\{B \to C, A \to B\}$ $\Rightarrow$ ∴ C is extraneous in $A \to BC$

#e.g., Consider the FD set

$F = \{A \to BC,\ B \to C\} \equiv \{A \to B,\ A \to C,\ B \to C\}$

① Check if B is extraneous in $A \to B$

{i.e. Check if $A \to B$ is redundant or not}

(∴ obtain $(A)^+$ wrt $(F - A \to B)$

i.e. $(A)^+$ wrt $\{A \to C,\ B \to C\} = \{A, C\} \Rightarrow B \notin (A)^+$ wrt $(F - A \to B)$

∴ B is not extraneous

or

we can say

$A \to B$ is not redundant

#e.g., Consider the FD set

$F = \{A \to BC,\ B \to C\} \equiv \{A \to B,\ A \to C,\ B \to C\}$

(2) Check if C is extraneous in $A \to C$

{i.e. Check if $A \to C$ is redundant or not}

(∴ obtain $(A)^+$ w.r.t $(F - A \to C)$

i.e. $(A)^+$ w.r.t $\{A \to B,\ B \to C\} = \{A, B, C\} \Rightarrow C \in (A)^+$ w.r.t $(F - A \to C)$

∴ C is extraneous

or

we can say $A \to$ is redundant



Topic : Procedure to obtain minimal cover of FD set

Step-1. Simplify RHS of all FDs (i.e., split the FDs such that RHS contain exactly one attribute)

Step-2. For all FDs find redundant (extraneous) attribute in LHS. { if any then remove that extaneous attribute }

Step-3. Eliminate all redundant FDs { i.e. check for extraneous attribute in RHS if any then remove it }

Step-4. Apply Union if needed

5. The result is minimal Cover

#e.g., Consider the following FD set

F ={AC →G

D→EG

BC→D

CG→BD

ACD→B

CE→AG

}

Find minimal cover of F.

Step-1

F = { AC → G
D → E
D → G
BC → D
CG → B
CG → D
ACD → B
CE → A
CE → G

#e.g., Consider the following FD set

$F = \{AC \to G$

$D \to EG$

$BC \to D$

$CG \to BD$

$ACD \to B$

$CE \to AG$

$\}$

Find minimal cover of F.

Step-1

$F = \{AC \to G,\ D \to E,\ D \to G,\ BC \to D,\ CG \to B,\ CG \to D,\ ACD \to B,\ CE \to A,\ CE \to G\}$

$(AC)^+ = \{A, C, G, D, \ldots\}$

D is extraneous w.r.t (AC)

$\therefore AC \to B$

Step-2

Check extraneous attribute in LHS

$AC \to G$:

$(A)^+ = \{A\}$ ∴ C is not extra

$(C)^+ = \{C\}$ ∴ A is not extra

$\Rightarrow AC \to G$

$D \to E$: Only one attribute in LHS ∴ Nothing in LHS can be extraneous

$\Rightarrow D \to E$

$D \to G$: =

$\Rightarrow D \to G$

$BC \to D$:

$(B)^+ = \{B\}$ ∴ C is not extraneous

$(C)^+ = \{C\}$ ∴ B is not extraneous

$\Rightarrow BC \to D$

$CG \to B$:

$(C)^+ = C$, $(G)^+ = G$: None is extraneous

$\Rightarrow CG \to B$

$CG \to D$

$(CD)^+ = \{C, D, E, G, B, A, \ldots\}$

∴ A is extraneous w.r.t (CD)

$\therefore CD \to B$

$(C)^+ = \{C\}$

$(D)^+ = \{D, E, G\}$

$CE \to A$:

$(C)^+ = \{C\}$, $(E)^+ = \{E\}$

∴ None is extraneous

$CE \to G$

At a time only one of this two will be used

If more than one option are available, then choose any one of them

#e.g., Consider the following FD set

$F = \{AC \rightarrow G$

$D \rightarrow EG$

$BC \rightarrow D$

$CG \rightarrow BD$

$ACD \rightarrow B$

$CE \rightarrow AG$

$\}$

Find minimal cover of F.

Step-1

$F = \{$

$AC \rightarrow G$

$D \rightarrow E$

$D \rightarrow G$

$BC \rightarrow D$

$CG \rightarrow B$

$CG \rightarrow D$

$ACD \rightarrow B$

$CE \rightarrow A$

$CE \rightarrow G$

$(AC)^+ = \{A, C, G, D, \ldots\}$

D is extraneous w.r.t. (AC)

$\therefore AC \rightarrow B$

Result of Step-2

$AC \rightarrow G$

$D \rightarrow E$

$D \rightarrow G$

$BC \rightarrow D$

$CG \rightarrow B$

$CG \rightarrow D$

$ACD \rightarrow B$

$CE \rightarrow A$

$CE \rightarrow G$

Step (3)

Not Redundant $\{(AC)^+ = \{A, C\}$

Not redundant $\{(D)^+ = \{D, G\}$

Not Redundant $\{(D)^+ = \{D, E\}$

Not Redundant $\{(BC)^+ = \{B, C\}$

$\{(CG)^+ = \{C, G, D, B, \ldots\}$

$\therefore CG \rightarrow B$ redundant

Remove → (will not be used in future)

$(CG)^+$ without $= \{C, G\}$

$\therefore$ Not Redundant

$(CD)^+ = \{C, D, E, G, A\}$ Not Redundant

$(CE)^+ = \{C, E, G, D, B\}$ $\therefore$ Not Redundant

$(CE)^+ = \{C, E, A, G, \ldots\}$

$\therefore CE \rightarrow G$ is Redundant

At a time only one of this two will be used

If more than one option are available, then choose any one of them

#e.g., Consider the following FD set

$F = \{AC \rightarrow G$
$D \rightarrow EG$
$BC \rightarrow D$
$CG \rightarrow BD$
$ACD \rightarrow B$
$CE \rightarrow AG$
$\}$

Find minimal cover of F.

Step-1

Step-4 Union

$AC \rightarrow G$
$D \rightarrow EG$
$BC \rightarrow D$
$CG \rightarrow BD$
$ACD \rightarrow B$
$CE \rightarrow AG$

$F =$ { $AC \rightarrow G$, $D \rightarrow E$, $D \rightarrow G$, $BC \rightarrow D$, $CG \rightarrow B$, $CG \rightarrow D$, $ACD \rightarrow B$, $CE \rightarrow A$, $CE \rightarrow G$ }

Result of Step-2

$AC \rightarrow G$
$D \rightarrow E$
$D \rightarrow G$
$BC \rightarrow D$
$CG \rightarrow B$
$CG \rightarrow D$
$ACD \rightarrow B$
$CE \rightarrow A$
$CE \rightarrow G$

Step ③

Not Redundant $\{(AC)^+ = \{A, C\}$
Not redundant $\{(D)^+ = \{D, G\}$
Not Redundant $\{(D)^+ = \{D, E\}$
Not Redundant $\{(BC)^+ = \{B, C\}$
$\{(CG)^+ = \{C, G, D, B, \ldots\}$

$\therefore CG \rightarrow B$ redundant
Remove → (will not be used in future)

$(CG)^+$ without $= \{C, G\}$
$\therefore$ Not Redundant

$(CD)^+ = \{C, D, E, G, A\}$ Not Redundant

$(CE)^+ = \{C, E, G, D, B\}$ $\therefore$ Not Redundant

$(CE)^+ = \{C, E, A, G, \ldots\}$

$\therefore CE \rightarrow G$ is Redundant

#e.g., Consider the following FD set

$F = \{A \to BC$

$CD \to E$

$E \to C$

$D \to AEH$

$ABH \to BD$

$DH \to BC$

$\}$

Find minimal cover of F.

① Step-1

$A \to B$
$A \to C$
$CD \to E$
$E \to C$
$D \to A$
$D \to E$
$D \to H$
~~$ABH \to B$~~ Trivial
$ABH \to D$
$DH \to B$
$DH \to C$

Step-2

$A \to B$
$A \to C$
~~$D \to E$~~
$E \to C$
$D \to A$
$D \to E$
$D \to H$

$(BH)^+ = \{B, H\}$ ∴ A is not extra
$(AH)^+ = \{A, H, B, C\}$ ∴ B is not extra
$(AB)^+ = \{A, B, C\}$ ∴ H is not extra

$ABH \to D$

~~$D \to B$~~
~~$D \to C$~~

Step-3

$A \to B$
$A \to C$
$E \to C$
$D \to A$
$D \to E$
$D \to H$

Step-4

$A \to BC$
$E \to C$
$D \to AEH$
$AH \to D$

$AH \to D$



Topic : NOTE

Minimal cover of FD set F need not be unique, but all minimal cover are logically equivalent.

* i.e. Fm_1 & Fm_2 are two minimal covers of F,

then $Fm_1 = Fm_2 = F$

#e.g., Consider the FD set

$F = \{AB \rightarrow C,\ B \rightarrow A,\ A \rightarrow B\}$

If we use $B \rightarrow A$ then A becomes extraneous in $AB \rightarrow C$

If we use $A \rightarrow B$, then B becomes extraneous in $AB \rightarrow C$



2 mins Summary

Topic Minimal Cover (Canonical Cover)

Topic Procedure to obtain minimal cover for a given FD set

Topic Schema refinement (Normalization)

Topic Problems because of redundancy in a relation


THANK - YOU
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